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Case Study

Transparent ONT Bacterial Genome Assembly

A reproducible, educational, and logic-first case study on bacterial genomics from first principles.

In genomics, a workflow is never just a chain of commands. It is a chain of biological and computational assumptions. If you do not understand the fundamentals behind each step, automation can hide errors instead of solving them.

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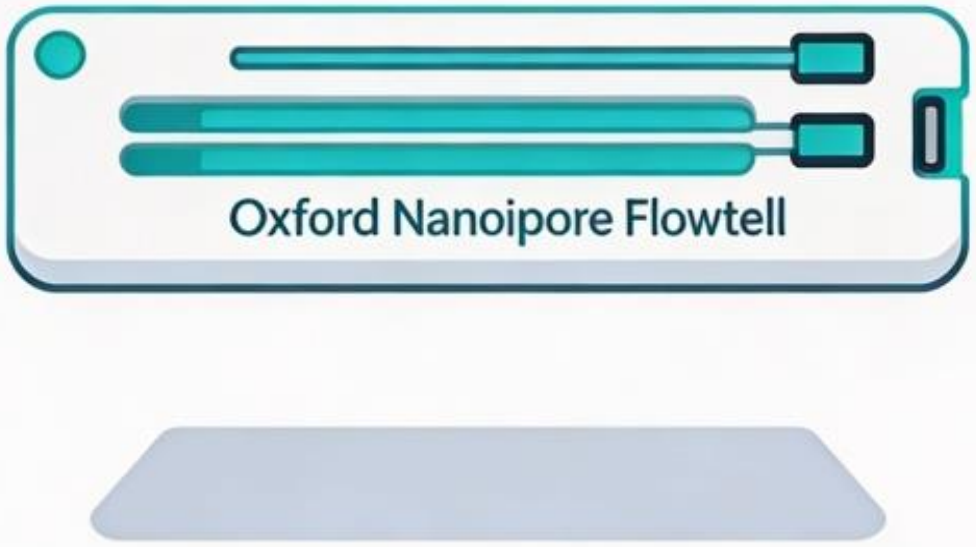
The Problem

Raw ONT reads are not yet a genome

Sequencing produces fragmented reads that must be computationally reconstructed.

Because different assemblers can yield divergent results, rigorous evaluation is required before any biological interpretation can be trusted.

This case study focuses first on reasoning, interpretation, and reproducibility before any one-click pipeline thinking.



Ancestra Genomics.

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Raw Reads QA

```
## Raw reads check (seqkit)
**Command:** seqkit stats -a raw/barcode07.fq
```
file   format type num_seqs   sum_len min_len avg_len max_len   Q1   Q2   Q3 sum_gap   N50 N50_num Q20(%) Q30(%) AvgQual GC(%) sum_n
raw/barcode07.fq FASTQ   DNA   37,054 392,624,046   462   10,596 165,410 3,525 4,294 10,991   0 24,577   4,324   83.98   57.02   16.96 41.29   0
```
num_seqs = 37,054
sum_len  = 392,624,046 bp  (~392.6 Mb)
avg_len  = 10,596 bp
N50      = 24,577 bp
Q20(%)   = 83.98
Q30(%)   = 57.02
GC(%)    = 41.29
```
```

# Running The Assemblers

```
Flye assembly
Command: flye --nano-hq raw/barcode07.fq --out-dir assemblies/flye --threads $(nproc)
```
total 9.3M
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 20:50 00-assembly
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 20:54 10-consensus
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 20:55 20-repeat
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 20:55 30-contigger
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 21:01 40-polishing
-rw-r--r-- 1 abane_ashu abane_ashu 3.7M Aug 12 21:01 assembly.fasta
-rw-r--r-- 1 abane_ashu abane_ashu 5.5M Aug 12 21:01 assembly_graph.gfa
-rw-r--r-- 1 abane_ashu abane_ashu 1.5K Aug 12 21:01 assembly_graph.gv
-rw-r--r-- 1 abane_ashu abane_ashu  120 Aug 12 21:01 assembly_info.txt
-rw-r--r-- 1 abane_ashu abane_ashu  45K Aug 12 21:01 flye.log
-rw-r--r-- 1 abane_ashu abane_ashu   92 Aug 12 21:01 params.json
```
Contigs & stats
```
contigs: 2
total_len = 3,780,699 bp
N50       = 3,773,355 bp
largest   = 3,773,355 bp
mean_cov  = 85×
```

```
## Raven assembly
**Command:**
```
raven \
  --threads $(nproc) \
  --graph assemblies/raven/raven.gfa \
  raw/barcode07.fq \
  > assemblies/raven/raven.fasta \
  2> logs/raven_withgraph_$(date +%F_%H%M).log
```

**Output files:**
```
total 15M
-rw-r--r-- 1 abane_ashu abane_ashu 6.2M Aug 12 21:51 raven.fasta
-rw-r--r-- 1 abane_ashu abane_ashu 7.8M Aug 12 21:51 raven.gfa
```

**Contigs & stats:**
**Command:**
```
grep -c '^>' assemblies/raven/raven.fasta
seqkit stats -a assemblies/raven/raven.fasta
```
**Results:**
```
contigs   = 28
sum_len   = 6,402,553 bp
min_len   = 32,563 bp
avg_len   = 228,663 bp
max_len   = 3,773,352 bp
N50   = 3,773,352 bp
GC(%)   = 46.73
```
```

Raw Reads Quality Check

SeqKit summary of ONT barcode reads and initial assembly snapshots from Flye, Raven, and Shasta for a reproducible evaluation

Metric	Value
Read count	37,054
Total bases	392.6 Mb
Average read length	10,596 bp
N50 (reads)	24,577 bp
Q20	83.98%
Q30	57.02%
GC (reads)	41.29%
Assembler	Contigs / Total length

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QC Metrics That Guide Assembly

Early QC reveals whether long-read yield and assembly choice will affect structure, completeness, and downstream interpretation

- Read yield and length distribution justify bacterial assembly if sufficient long reads present. 392.6 Mb for a bacterial genome strongly suggests usable coverage.
- Additionally, mean read length 10.6 kb and N50 24.6 kb indicate long reads that can help bridge repeats better than short reads
- Different assemblers Flye, Raven, and Shasta produced markedly different assembly structures from the same ONT input. Therefore DO NOT confuse good input with good assembly
- These early differences make downstream structural and completeness evaluation essential before biological interpretation.

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From Raw Reads To Biological Meaning

QA & Assembly

Raw read quality check via SeqKit followed by parallel assembly using Flye, Raven, and Shasta. These were arbitrary, chosen based on the type of reads being analysed

Multi-Metric Evaluation

Rigorous benchmarking using QUAST, KAT, BUSCO, and Bandage to assess structural and sequence integrity.

Taxonomic Context

16S rRNA extraction and phylogenetic analysis to confirm species identity and evolutionary placement.

Functional Annotation

Final metabolic profiling using Prokka and BlastKOALA to link structural data to biological function.

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No single metric is enough

QUAST

Measures assembly size and N50 to determine physical contiguity and length.

KAT

Analyzes k-mer distribution to ensure the assembly reflects the raw read content.

BUSCO

Evaluates the presence of universal single-copy orthologs to confirm biological completeness.

Bandage

Visualizes the underlying assembly graph to identify structural ambiguities or circularity.

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Assembly Evaluation Across Flye, Raven, and Shasta

QUAST, KAT, BUSCO, and Bandage together reveal which assembly is most trustworthy for downstream biological interpretation. Multiple evidence streams are required for a confident choice.

Metric	Flye	Raven	Shasta
Contigs	2	28	13
Total length	3,780,699 bp	6,402,553 bp	3,155,007 bp
N50	3,773,355 bp	3,773,352 bp	437,312 bp
BUSCO	98.4% complete, 0.0% duplicated, 0.0% fragmented, 1.6% missing	54.8% complete, 40.4% duplicated, 0.0% fragmented, 0.8% missing	78.2% complete, 0.0% duplicated, 5.7% fragmented, 16.1% missing
KAT	strongest agreement / most plausible completeness	suspicious excess / possible redundancy	weaker completeness support
Bandage graph	clean circular graph with small accessory structure	more unresolved or branched structure	fragmented graph structure
Overall interpretation	strongest overall candidate	likely over-assembly or redundant reconstruction	more fragmented and less reliable for downstream analysis

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Interpretation

From assembly to biological meaning

Phylogenetic Context

- 16S rRNA extraction confirms species identity
- Tree analysis places the isolate within its family
- Validation of evolutionary conserved regions

Functional Annotation

- Prokka identifies genes and open reading frames
- BlastKOALA maps metabolic pathways
- Flye yielded the richest functional profiles

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16S Identification and Phylogenetic Context

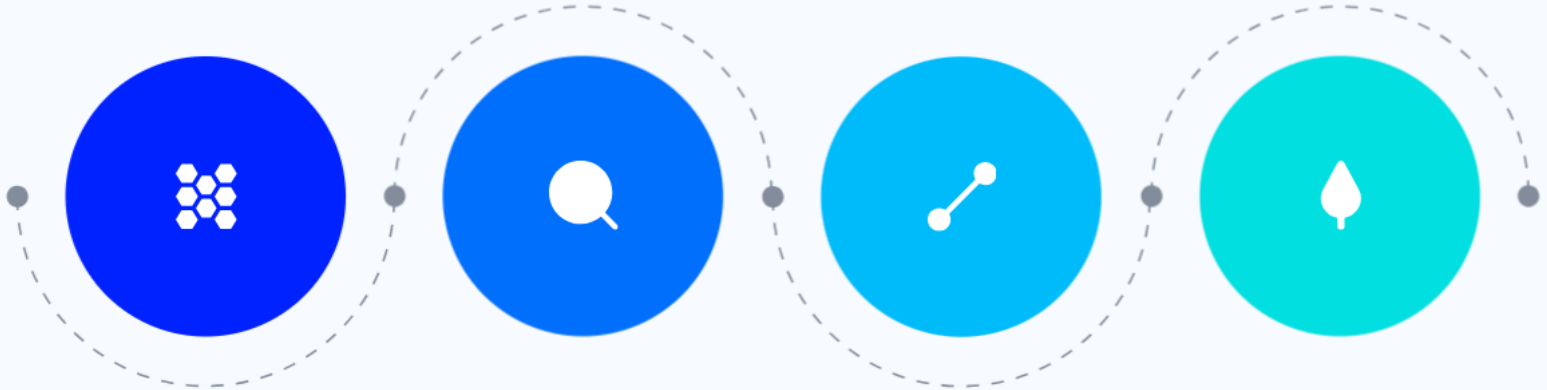
Extracted 16S rRNA sequences were compared to bacterial references and placed in phylogenetic context to verify taxonomic identity before deeper interpretation.

16S Extraction

Extract 16S rRNA sequences from genome assemblies for marker identification.

Multi-Alignment

Align sample sequences with reference homologs for phylogenetic inference.



BLAST Search

Query 16S sequences against curated bacterial databases for top matches.

Tree Inference

Build phylogenetic tree to assess evolutionary placement and species context.

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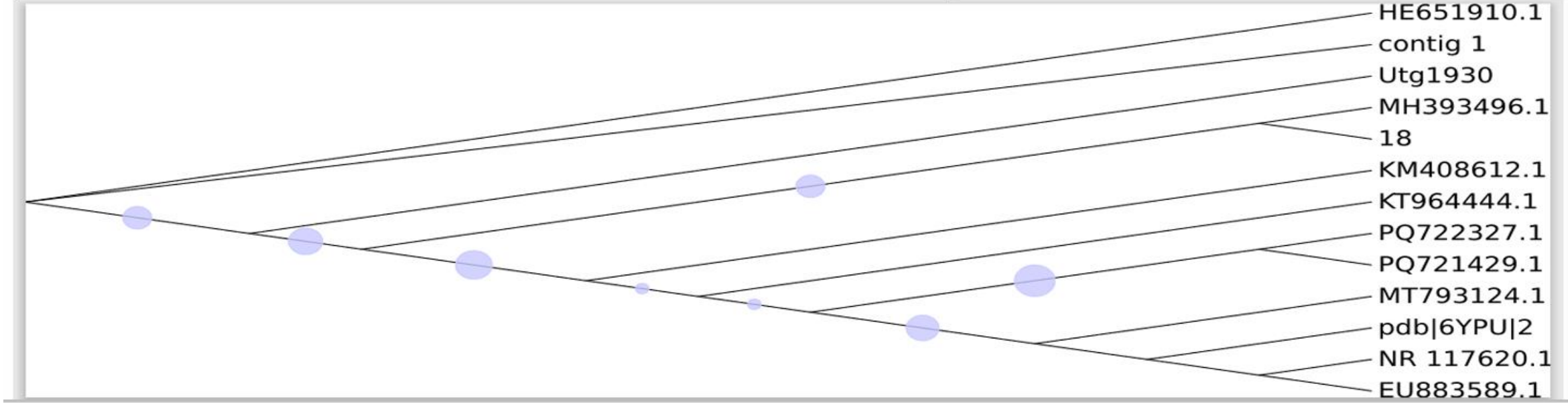
Phylogenetic Tree Analysis and Comparison

```
GNU nano 7.2                                                                    WORKFLOW.md *
 This allows quick inspection of species names, accession numbers, and % identity.

Output:
```text
==> flye_blastn.tsv <==
contig_1:1206335-1207868(+)     HE651910        100.000 1529    1533    100     2824    Acinetobacter baumannii 16S rRNA gene, strain 16842
contig_1:1206335-1207868(+)     6YPU_2  99.870  1533    1533    100     2820    Chain 2, 16S ribosomal RNA
contig_1:1206335-1207868(+)     MT793124        99.870  1533    1533    100     2820    Acinetobacter baumannii strain AB1 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     NR_117620       99.869  1529    1533    100     2813    Acinetobacter baumannii strain ATCC 19606 16S ribosomal RNA, partial sequence
contig_1:1206335-1207868(+)     EU883589        99.869  1529    1533    100     2813    Acinetobacter baumannii strain AIIMS 11 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     KT964444        99.804  1529    1533    100     2808    Acinetobacter baumannii strain MHHB02 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     KM408612        99.804  1529    1533    100     2808    Acinetobacter baumannii strain L317 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     MH393496        99.804  1528    1533    100     2806    Acinetobacter baumannii strain FOP220 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     PQ722327        99.869  1525    1533    99      2806    Acinetobacter baumannii strain Aci_bau1 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     PQ721429        99.869  1525    1533    99      2806    Acinetobacter baumannii strain Ac5 16S ribosomal RNA gene, partial sequence

==> raven_blastn.tsv <==
Utg1930:37498-39035(-)  MN429133        99.610  1537    1537    100     2804    Klebsiella pneumoniae strain NUBRI-K 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  PQ865839        99.610  1537    1537    100     2804    Uncultured Klebsiella sp. clone K-pneumoniae-1 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  PP140960        99.610  1537    1537    100     2804    Klebsiella sp. strain SG01 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  AB642255        99.674  1533    1537    100     2802    Klebsiella pneumoniae gene for 16S rRNA, partial sequence
Utg1930:37498-39035(-)  MZ401370        99.545  1537    1537    100     2798    Klebsiella pneumoniae strain M27 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  MZ401365        99.545  1537    1537    100     2798    Klebsiella pneumoniae strain M3 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  JN644536        99.738  1526    1537    99      2795    Klebsiella pneumoniae strain BR10_2A 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  MZ401368        99.480  1537    1537    100     2793    Klebsiella pneumoniae strain M20 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  MZ401364        99.480  1537    1537    100     2793    Klebsiella pneumoniae strain M2 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  AB642256        99.543  1533    1537    100     2791    Klebsiella pneumoniae gene for 16S rRNA, partial sequence

==> shasta_blastn.tsv <==
18:21581-23115(+)       HE651910        99.804  1530    1534    100     2808    Acinetobacter baumannii 16S rRNA gene, strain 16842
18:21581-23115(+)       6YPU_2  99.674  1534    1534    100     2804    Chain 2, 16S ribosomal RNA
18:21581-23115(+)       MT793124        99.674  1534    1534    100     2804    Acinetobacter baumannii strain AB1 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       KM408612        99.739  1530    1534    100     2802    Acinetobacter baumannii strain L317 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       MH393496        99.738  1529    1534    100     2800    Acinetobacter baumannii strain FOP220 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       EU078177        99.738  1529    1534    100     2800    Acinetobacter baumannii strain BD5 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       NR_117620       99.673  1530    1534    100     2796    Acinetobacter baumannii strain ATCC 19606 16S ribosomal RNA, partial sequence
18:21581-23115(+)       EU883589        99.673  1530    1534    100     2796    Acinetobacter baumannii strain AIIMS 11 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       KT964444        99.608  1530    1534    100     2791    Acinetobacter baumannii strain MHHB02 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       PQ722327        99.672  1526    1534    99      2789    Acinetobacter baumannii strain Aci_bau1 16S ribosomal RNA gene, partial sequence
```


```

Functional Annotation and Metabolic Interpretation

After selecting the strongest assembly, Prokka and BlastKOALA were used to translate assembled sequence into gene-level annotation and pathway-level metabolic context for evidence-based biological interpretation.

Genome Select

Identify and select the most credible reconstructed genome assembly to serve as the reliable foundation for all downstream annotation and analysis.

Prokka Annotate

Use Prokka to generate standardized gene and protein features, creating a structured and consistent annotation layer for the assembly.

Export Models

Collect predicted protein and gene models in standardized formats to facilitate downstream functional assignment and comparative mapping tasks.

KEGG Mapping

Map protein sequences to KEGG orthology using BlastKOALA to define metabolic and functional context, highlighting core survival and stress systems.

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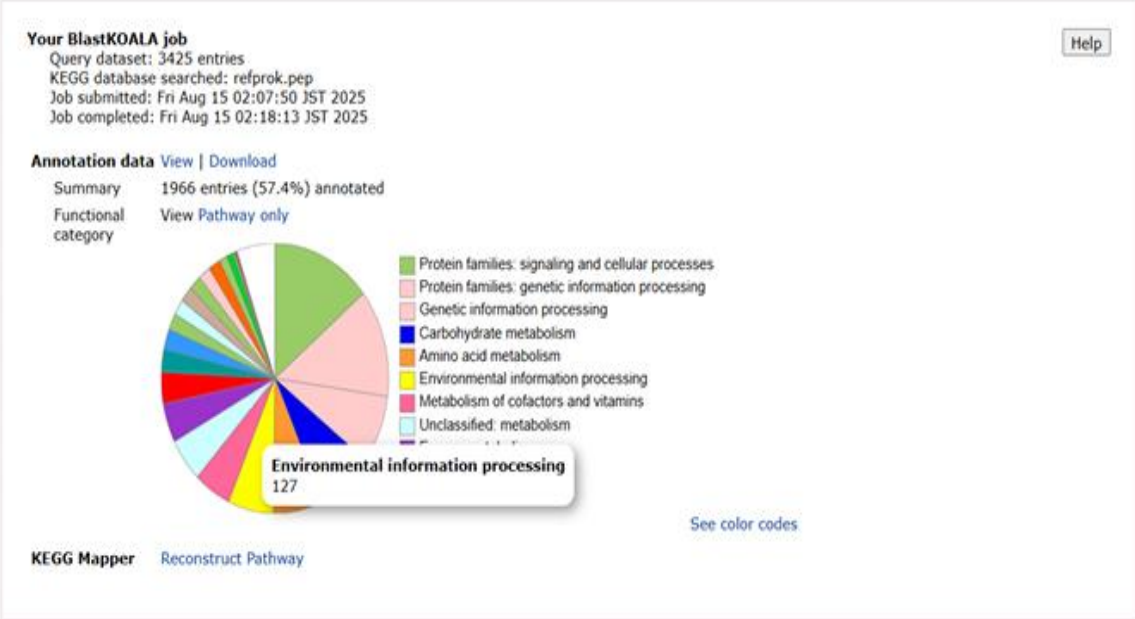

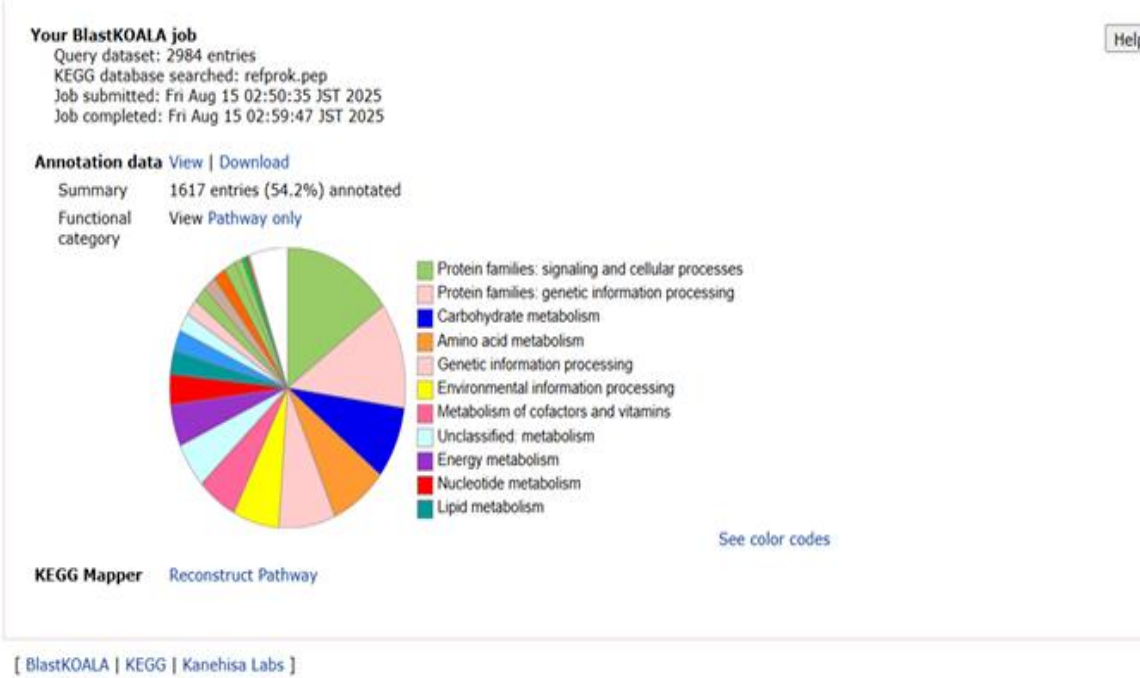

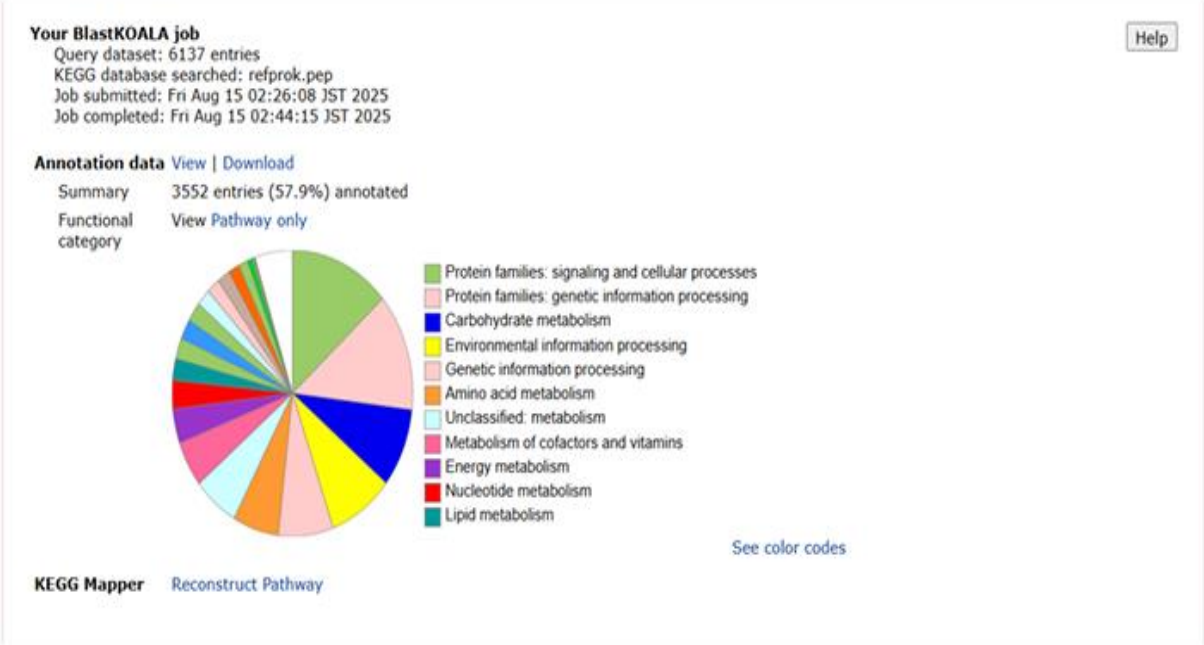

Fig A: Acinobacter baumanii (FLYE)- proteins predicted is 57.5%

Fig B: Klebsiella pneumoniae (RAVEN)- Proteins predicted is 57.9%

Fig C: Acinobacter baumanii (SHASTA)- Proteins predicted is 54.2%

Benchmarking

Compare, don’t assume

Assembler	Contiguity	Completeness	Metabolic Prediction	Verdict
Flye	High	Most Complete	57.5%	Gold Standard
Raven	Moderate	High (KAT/BUSCO)	57.9%	Divergent Phylogeny
Shasta	Fragmented	Incomplete	54.2%	Conserved Regions Only

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Understanding before automation

Confidence in genomics is built through transparent reasoning and comparison.

The value of this workflow lies not in automation alone, but in making every decision from raw reads to biological interpretation transparent and defensible.

In genomics, trustworthy results come from understanding the process deeply enough to question, compare, and justify what you carry forward.

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